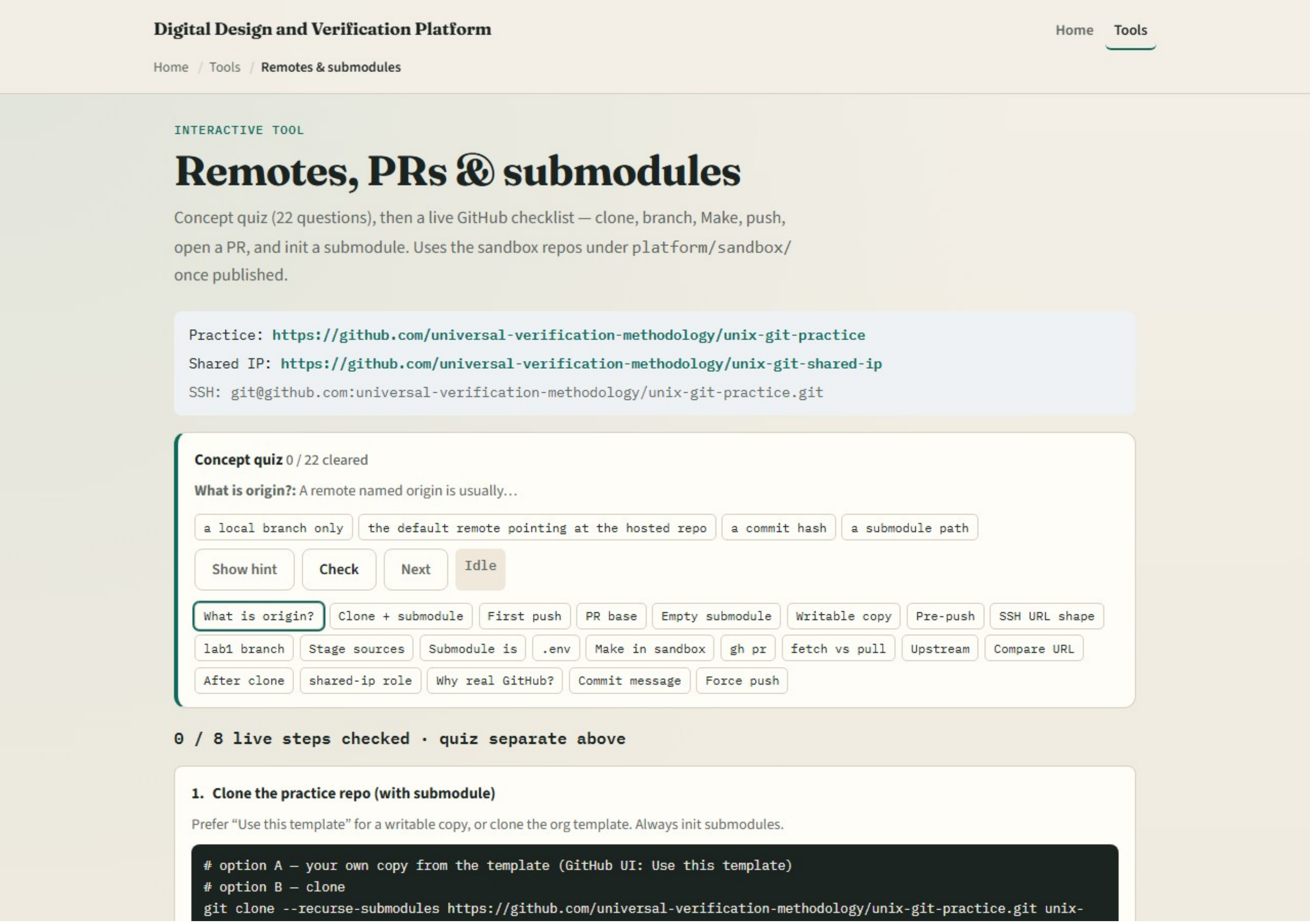
# Remotes, PRs & submodules

Local Git is only half the story

# Push, review, embed

- Add a remote with a URL
- Reviewers read the diff, you push more commits to the same branch if needed
- A submodule records a gitlink, the path plus an exact commit, not a casual copy of files
- After cloning a repo with submodules, you must initialize them or those folders stay empty

# Browser lab




Remotes lab — quiz and live checklist

# Real Git practice

```
wsl — module15-remotes/examples/team_workflow

# Track A — real Unix
# real shell session (Track A)

# practice sandbox: /tmp/git-remotes-ObbWLg

$ git init

$ echo "practice v1" > README.md

$ git add README.md

$ git commit -m "init"

$ git branch -M main

$ git init --bare ../origin.git

$ git remote add origin ../origin.git

$ git push -u origin main

$ git checkout -b lab1

$ echo "lab1 change" >> README.md

$ git add README.md

$ git commit -m "lab1: update README"

$ git push -u origin lab1

$ git remote -v
origin  /tmp/git-remotes-ObbWLg/origin.git (fetch)
origin  /tmp/git-remotes-ObbWLg/origin.git (push)

$ git submodule add ../shared-lib external/shared-ip

$ git submodule status
 849c484fba19376bdcf1d8dc322f9212ee618202 external/shared-ip (heads/master)
```

Real shell — remote, push, submodule

# Real Git practice — try these

```
# git init — create a new repository here
git init

# echo … > README.md — first commit on default branch
echo "practice v1" > README.md

git add README.md

git commit -m "init"

# git branch -M main — name the default branch main
git branch -M main

# git init --bare ../origin.git — bare remote (simulates GitHub)
git init --bare ../origin.git
```

# Pitfalls to watch

- Do not force-push shared default branches like main
- After clone, run submodule init if embedded folders are empty
- A pull request is for review and visibility
- And remember

# Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, clear a few quiz items and read the live GitHub steps end to end
- On real Git, practice push with upstream and inspect a submodule status line
- When you are ready